

Phylogenetic signal in subgenus *Rubus* (bramble, blackberry) leaf shape using Geometric Morphometrics and Elliptic Fourier Analysis

Juniper B. Kiss¹, Dániel G. Knapp², Gábor M. Kovács², Michal Sochor³, Nigel Cooper⁴

INTRODUCTION

Plant phenotypic plasticity and different ways of genetic recombination during clonal and sexual reproduction make the identification of some plant species difficult. Although DNA barcoding has revolutionised species identification, polyploidy, hybridisation and apomixis pose challenges to this process.

The genus *Rubus* consists of many commercially important species while subgenus *Rubus* (brambles) is ecologically important and known to include noxious and highly invasive environmental weeds worldwide. Brambles are also known to be taxonomically challenging. They have been described as biotypes, morphotypes, species and micro-species (Edees et al., 1988).

The complexity of their classification is mainly caused by (i) nature of polyploidy (Sochor et al. 2015), (ii) variable reproduction

(sexual, asexual, rapid hybridization; Šarhanová et al., 2012) and (iii) geographic parthenogenesis (Krutto et al., 2010).

In their morphology-based classification system, the micro-species are grouped into sections, subsections, and series. Their identification focuses on characters of their leaves, prickles, flowers, and root systems.

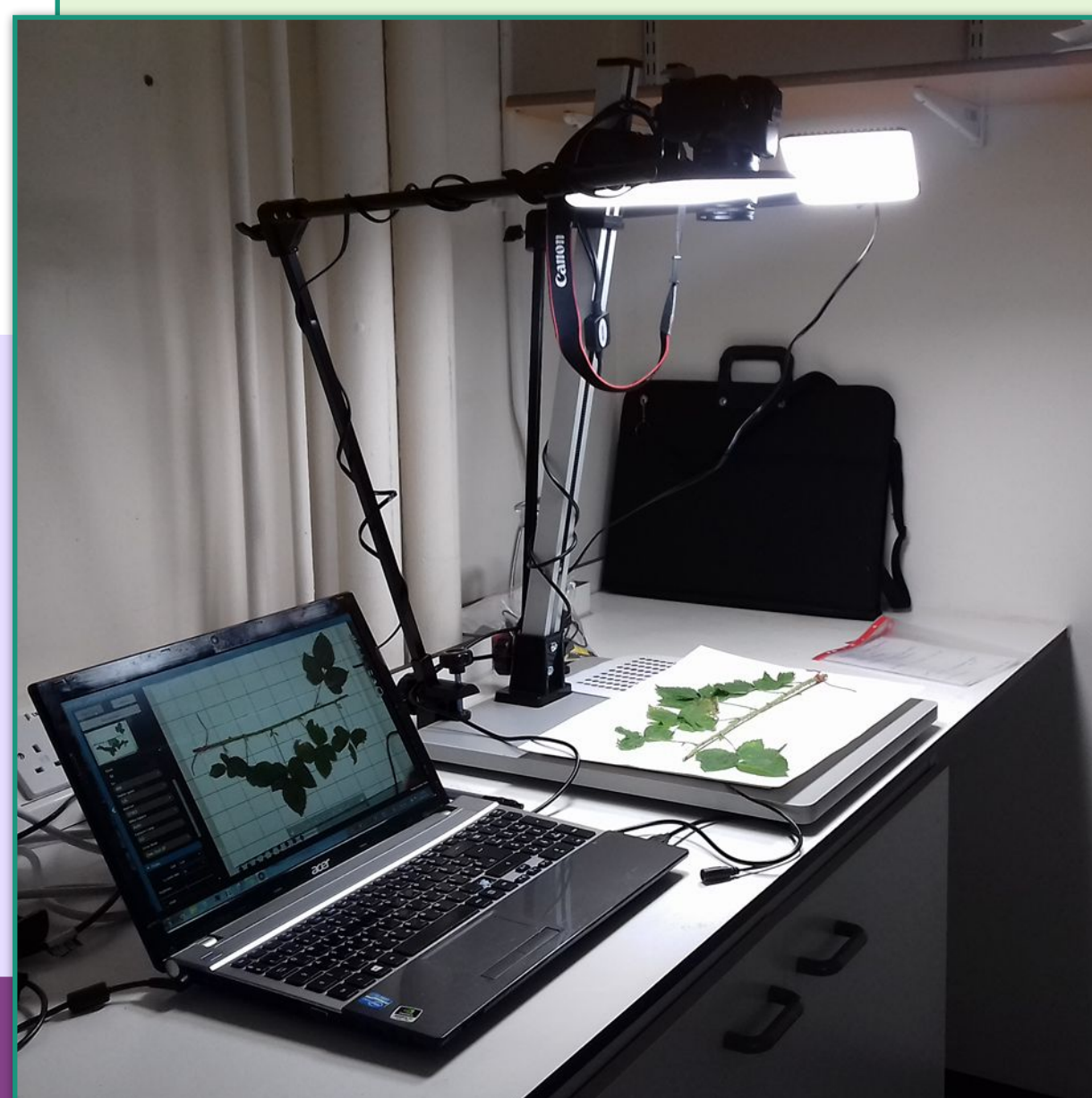
Our hypothesis was that same haplotypes will have different shapes in different environments. We suggest that leaf shape within a series is an unreliable identification character.

Here, we combined image analysis with molecular analysis to explore the compound leaf morphology in different environments of bramble haplotypes in the UK.

SAMPLES AND METHODS

Sample collection and imaging

- Specimens were collected in the UK in Woolacombe (WC, sandy beach), Comb Martin (CM, coast with clay), Wells-next-the-Sea (WS, saltmarsh), Holkham (HH, sandy beach), and Thetford Forest (TF, pine forest).
- Samples were identified down to series level.
- Cane number was recorded in a 1 m² grid around the sampled cane.
- Coverage was estimated by using a spherical densiometer (Forestry Suppliers).
- Bramble canes with leaves attached were cut and pressed in the field.
- DSLR camera (Canon 500D) was mounted on copy stand with LED lights (SG CS-720, Speed Graphic). Camera was remotely controlled with DigiCam, for uniform imaging.



Geometric morphometrics

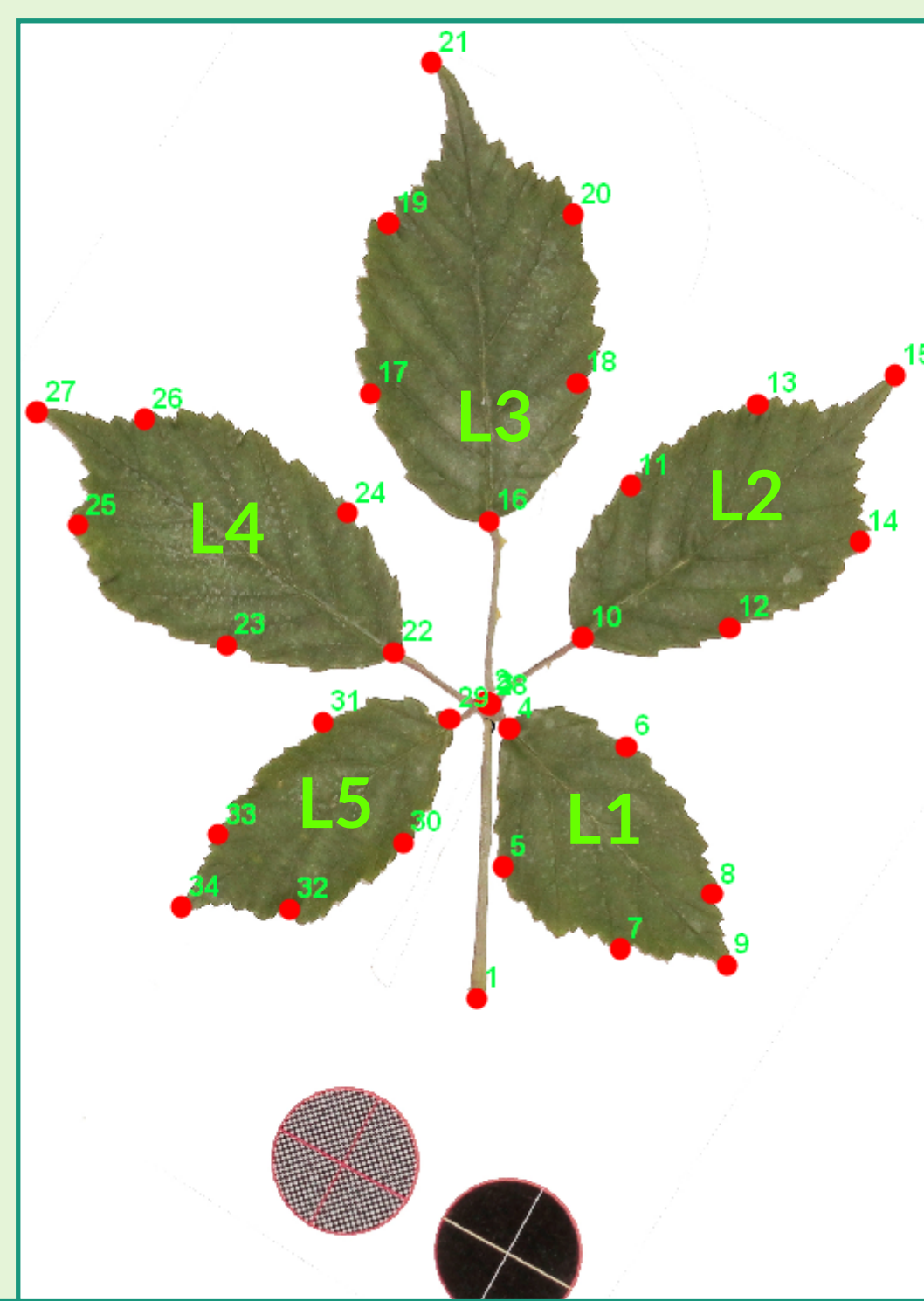
- Two five-foliate leaves were used for 115 specimens.
- TpsUtil software was used for building tps file from chosen images.
- TpsDig 1.4 (Rholf, 2004) was used for scaling and recording 34 landmarks on five-foliate leaves.
- For landmarks, capturing the leaf widths, ImageJ macro function was written for dividing the primary vein into four segments and were used to record the landmarks on the two sides of the leaves.
- The tps file was imported into MorphoJ (Klingenberg, 2011) software.
- Procrustes superimposition around the principal axes removed the effect of size on the morphological variation within the data set.

Tests used

- Principal Component Analysis (PCA)
- Canonical Variate Analysis (CVA, 10 000 permutations)
- Discrimination Analysis (DS, 10 000 permutations)
- Procrustes ANOVA (Klingenberg et al., 1998)
- Squared-change parsimony for phylogeny mapping (Klingenberg and Gidaszewski, 2010; 10 000 permutations)
- Regression

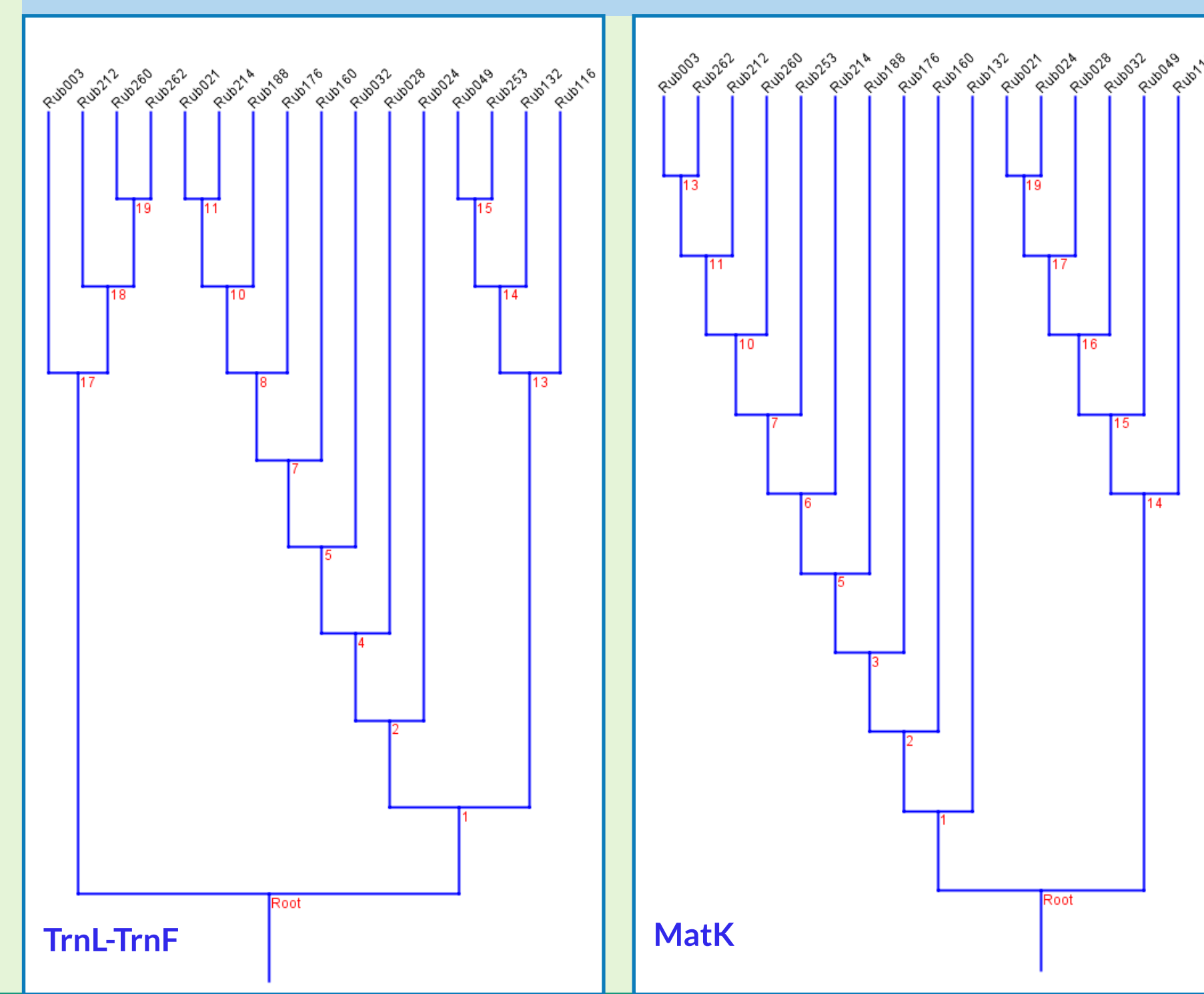
Elliptic Fourier Analysis

- Terminal leaves of all specimens (1456) were converted into binary bitmap files and imported into SHAPE v1.3 software package (Iwata and Ukai, 2002).
- The contour information was described by a chain code, which was normalised. The data was imported with 'Nef2Coe' function into Momocs R package (Bonhomme et al., 2014).
- PCA was used to visualise the distribution of leaf outline for each series.



Molecular Methods

- Twenty-three representative samples were selected for a three loci analysis.
- DNA was extracted from dried leaves using DNeasy Plant Mini Kit (Qiagen). Using DreamTaq polymerase (Thermo Scientific), ITS, maturase K (matK) and trnL-trnF regions were amplified with the primer pairs ITS1 and ITS4, XFA and AST, c and f.
- Sanger sequencing was carried out by LGC GmbH (Berlin, Germany).
- Sequences were aligned in BioEdit v7.0.5 (Hall, 1999). Haplotypes were identified according to Sochor et al. (2015). Maximum Likelihood tests were carried out in MEGA7 (Kumar et al., 2016) with 500 bootstrap sampling. Newick files containing the node lengths were imported in Mesquite 3.40 (Maddison and Maddison, 2018) to convert them into NEXUS files with bootstrap values.



RESULTS

Geometric morphometrics

PCA

PCA did not show a separation between series. PC1 explained 32% of the total variance and represented the terminal leaf elongation, petiole length and insertion point of basal leaflets. PC2 explained 14% variance and contained the main changes of the basal leaflets width, length and angle.

CVA and DS

CVA separated leaf morphology according to series ($P < .0001$). DS also confirmed the significance ($P < .001$). Within series, different locations, cane densities and coverage also separated shapes significantly ($P < .001$, few exceptions). DS was used to further investigate shape differences between sites which were significant when large amount of samples were available ($P < .05$).

Symmetry and Allometry

Symmetry was enforced on data and Procrustes ANOVA showed there were significant differences between series ($P < 0.0001$). Location and cane density were also significant effects on leaf shape ($P < .0001$) on the symmetric component. Pooled within-series regressions of leaf shape on log-transformed centroid size accounted for 2.48% of total variation and had a significant effect on shape ($P = 0.0004$).

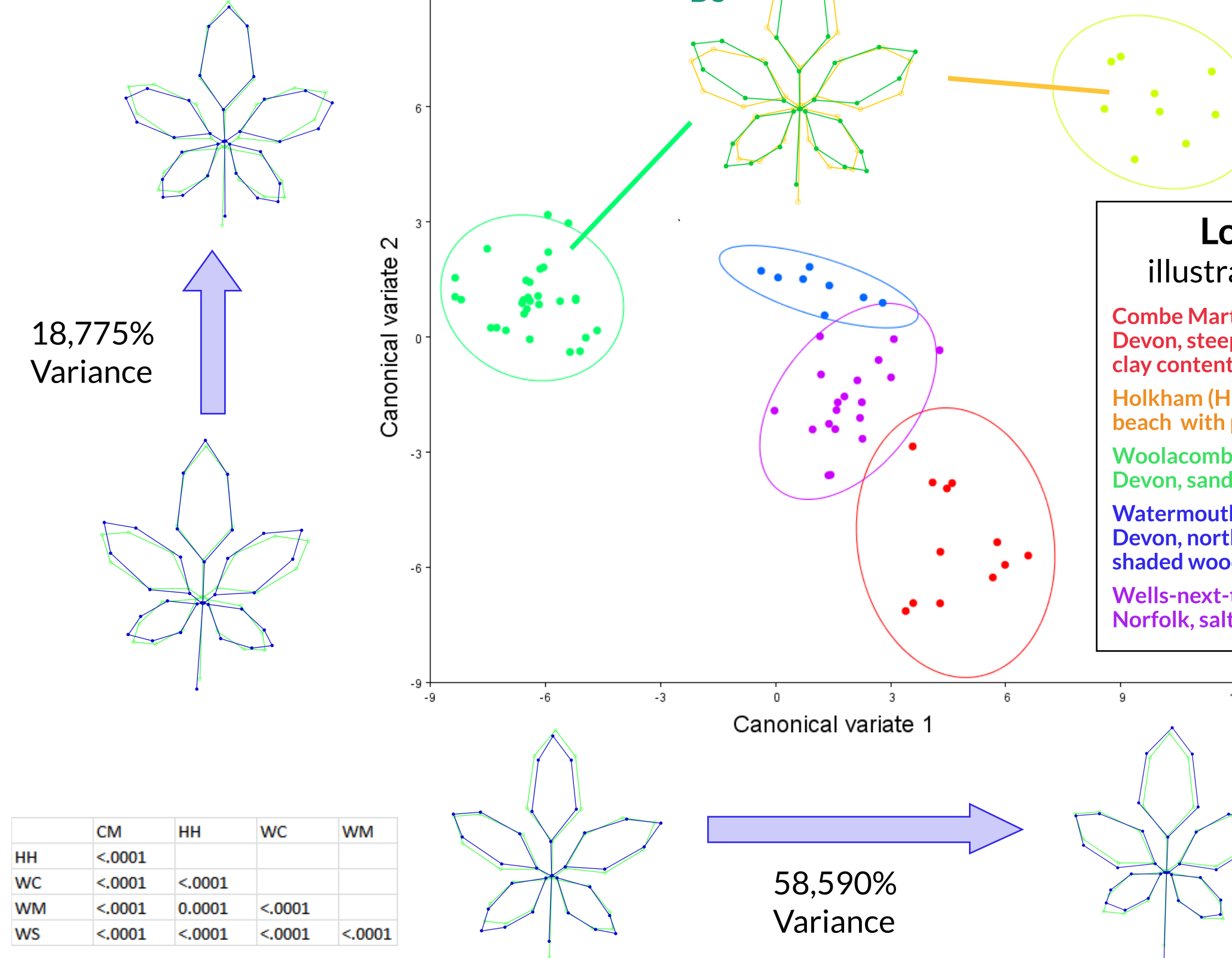
Projecting phylogeny into Shape Space

Both trnL-trnF ($P < .0001$) and matK regions ($P = 0.0131$) were significant when mapped onto the shape space, created by a within-series pooled CVA. Therefore, there was a significant phylogenetic signal within the leaf shapes. Haplotypes were separately projected into CVAs, pooled within locations and showed that same haplotypes have different shapes at different locations.

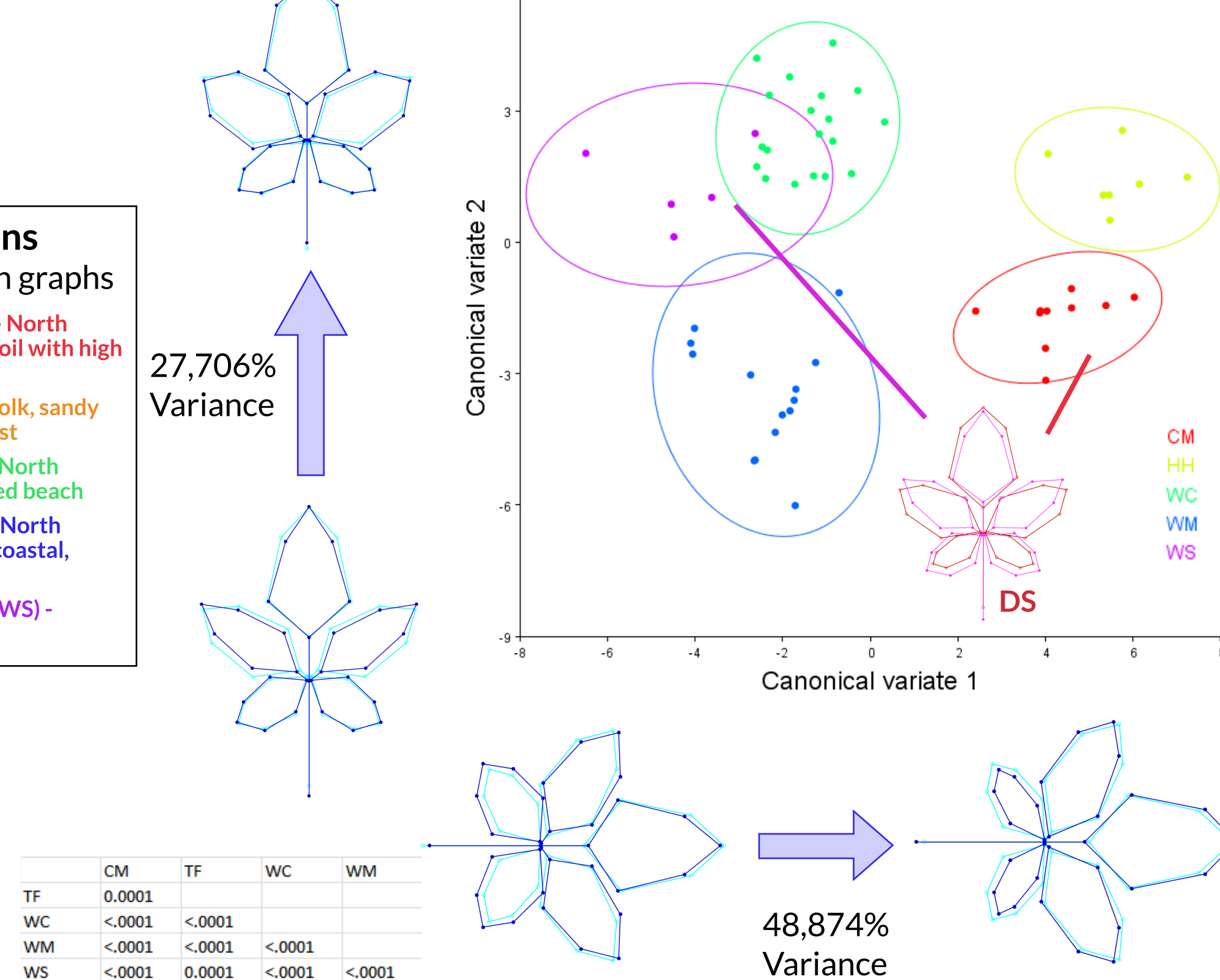
SUMMARY

Bramble morphology has not been found to be consistent with their phylogeny previously. Our data suggests, that between series, there is a phylogenetic signal but leaf morphology is not reliable for micro-species identification. This challenges the current bramble classification system.

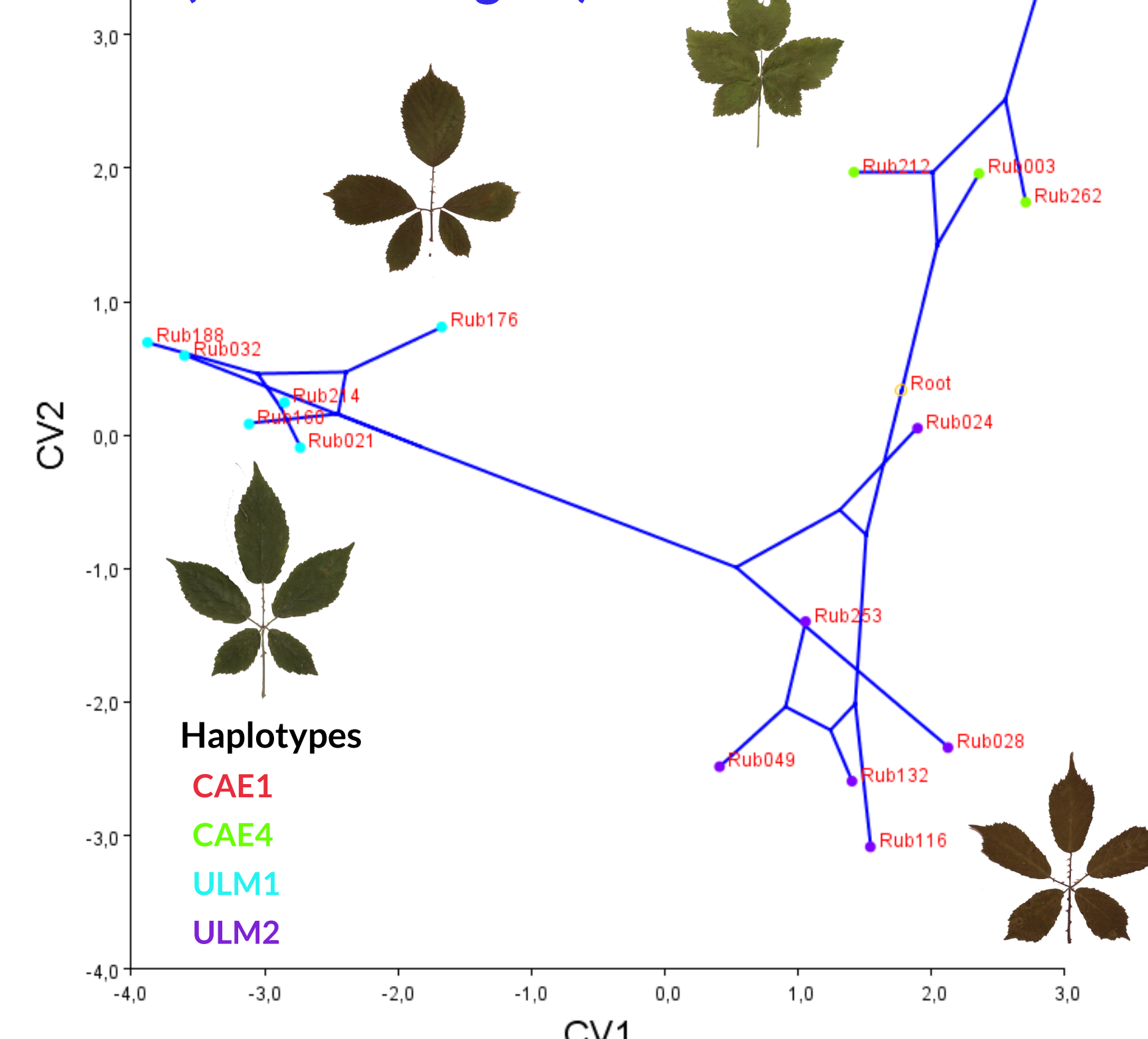
Series Sylvatici - Asymmetrical Component - Locations



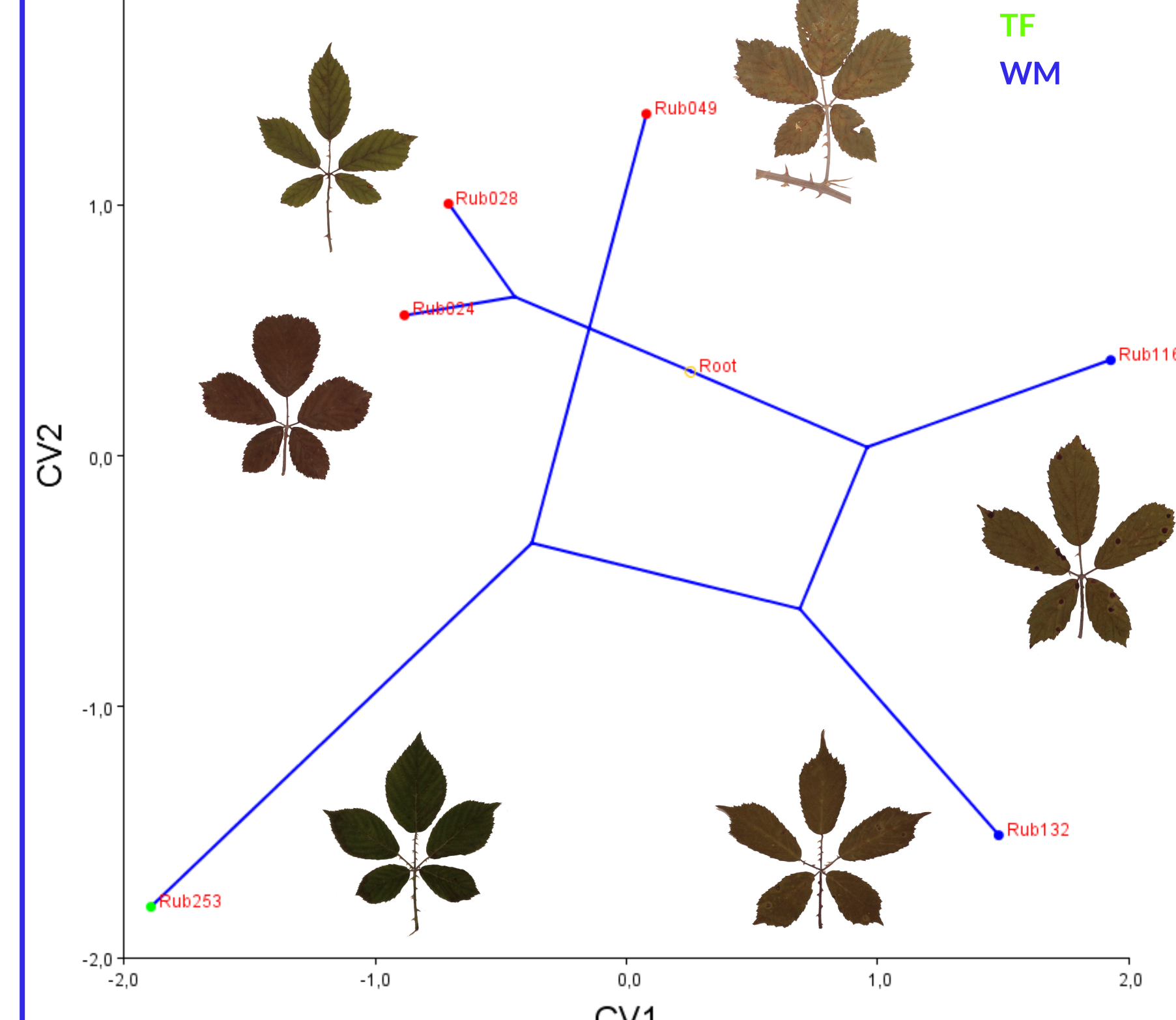
Series Corylifolii - Symmetrical Component - Locations



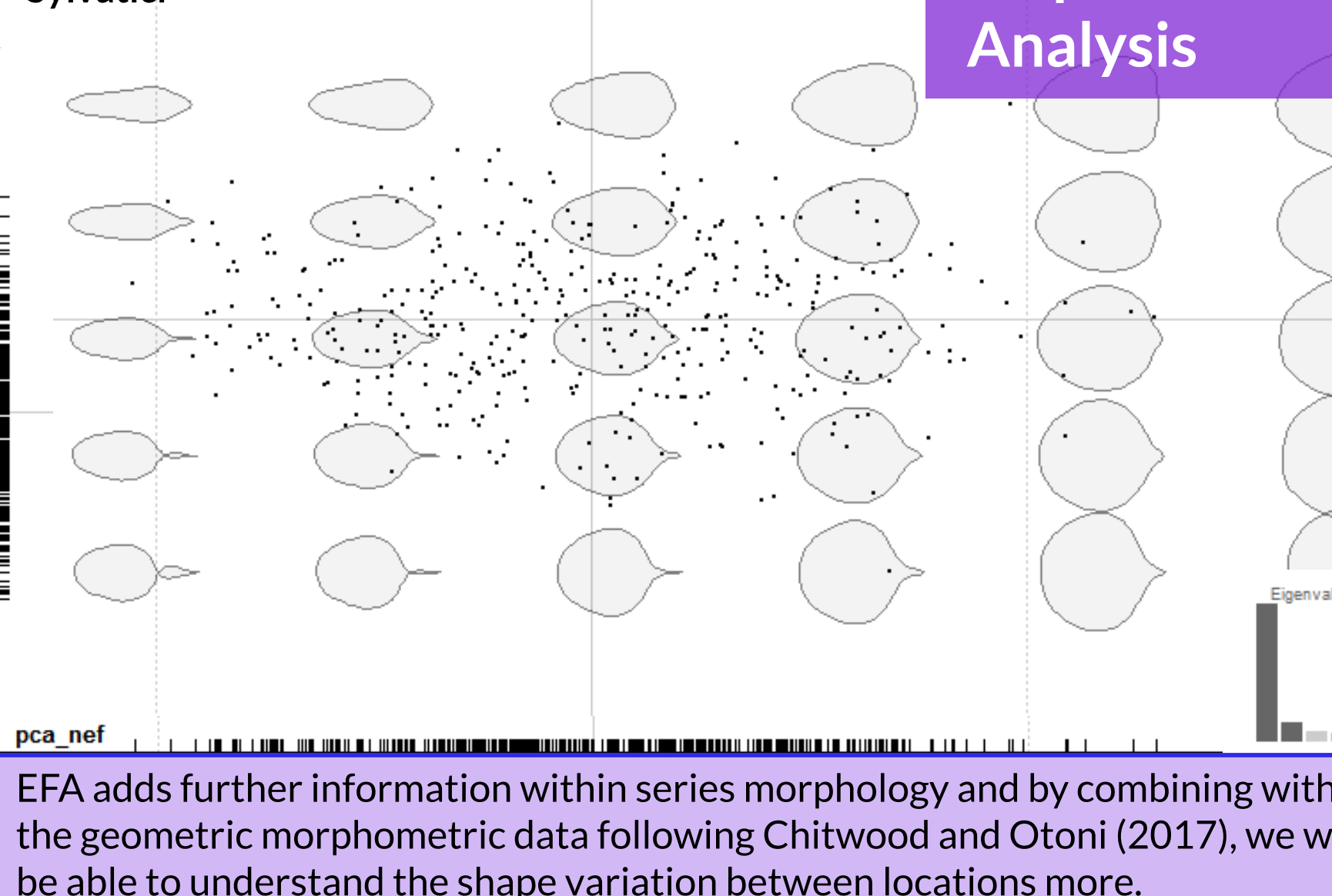
Haplotype shapes (trnL-trnF region)



ULM1 haplotype at different locations



Series Sylvatici



Elliptic Fourier Analysis

EFA adds further information within series morphology and by combining with the geometric morphometric data following Chitwood and Otoni (2017), we will be able to understand the shape variation between locations more.

ACKNOWLEDGEMENTS

Sample collection was supported by the John Ray Summer Research Grant. Lab space was provided by Anglia Ruskin University for imaging. Lab space and equipment for molecular analysis was provided by Eotvos Lorand University. We thank the recommendations of Philip Pugh for project design. Travel award to Phenome 2018 conference was provided by ARPA-E and ASPB.

REFERENCES

- Bonhomme, V., Pico, S., Gaucherel, C. and Claude, J., 2014. Momocs: outline analysis using R. *Journal of Statistical Software*, 54(13), pp.1-24.
- Chitwood, D.H. and Otoni, W.C., 2017. Morphometric analysis of Passiflora leaves: the relationship between landmarks of the vasculature and elliptical Fourier descriptors of the blade. *GigaScience*, 6(1), pp.1-13.
- Edees, E.S., Newton, A. and Kent, D.H., 1988. *Brambles of the British Isles*. Ray Society.
- Klingenberg, C.P. and Gidaszewski, N.A., 2010. Testing and quantifying phylogenetic signals and homoplasy in morphometric data. *Systematic biology*, 59(3), pp.245-261.
- Klingenberg, C.P., 2011. MorphoJ: an integrated software package for geometric morphometrics. *Molecular Ecology Resources*, 11(2), pp.353-357.
- Šarhanová, P., Vašut, R.J., Dančák, M., Bureš, P. and Trávníček, B., 2012. New insights into the variability of reproduction modes in European populations of *Rubus* subgen. *Rubus*: how sexual are polyploid brambles? Sexual plant reproduction, 25(4), pp.319-335.
- Sochor, M., Vašut, R.J., Šarhan, T.F. and Trávníček, B., 2015. How just a few makes a lot: speciation via reticulation and apomixis on example of European brambles (*Rubus* subgen. *Rubus*, Rosaceae). *Molecular Phylogenetics and Evolution*, 89, pp.13-27.